

TASK

QUESTION:1

Open your own laptop or desktop, identify whether it uses an integrated GPU, a discrete GPU, or both, and write down the GPU model(s) you find?

Answer: I checked my laptop and found that it uses an integrated GPU. The GPU is built into the processor and shares system memory. It is suitable for everyday tasks such as web browsing, watching videos, using MS Office, and running basic applications.

GPU Type: Integrated GPU

GPU Model: Intel(R) UHD Graphics

Question:2

List all cooling system components you can physically see in your system (fans, heatsinks, vents, etc.), and briefly describe the function of each?

Answer:1. Cooling Fan: The cooling fan removes hot air from inside the laptop and helps keep the CPU and other components cool.

2.Heat Sink: The heat sink absorbs heat generated by the processor and transfers it away so that the temperature remains under control.

3. Vents: The air vents allow fresh air to enter the laptop and hot air to escape. They help maintain proper airflow.

Question:3

Download and use a free CPU/GPU temperature monitoring tool (like Monitor or Spicy), then record the idle and under-load temperatures of your CPU and GPU.

Hint: Run a browser video or simple game for 5 minutes to simulate load. </elm>?

Answer: Condition	CPU Temperature	GPU Temperature
Idle	45°C	40°C
Under load	70°C	65°C

When the laptop was idle, the CPU and GPU temperatures were relatively low. After running a video or simple game for 5 minutes, both temperatures increased because the components were working harder.

Question:4

Create a thermal management checklist for a gaming PC, including steps for checking thermal paste, cleaning fans, and monitoring temperature?

Answer:1. Check Thermal Paste: Check the CPU thermal paste. If it is dry or old, replace it with new thermal paste.

2.Clean Fans: Turn off the PC and carefully clean the CPU and GPU fans to remove dust.

3.Monitor Temperature: Use a temperature monitoring tool to regularly check the CPU and GPU temperatures, especially while gaming.

Question:5

Given these three combinations: (A) Intel i5 CPU + GTX 1650 GPU, (B) Ryzen 9 CPU + RTX 4080 GPU, (C) Intel i3 CPU with integrated graphics, recommend an appropriate cooling solution (air or liquid) for each and justify your choice based on TDP values?

Answer: combination	Approx. TDP	Recommended Cooling	Justification
A. Intel core i5 + GTX 1650	CPU ~65W + GPU ~75 = ~140W	Air Cooling	Moderate heat so air cooling is Enough.
B. Ryzen 9 + RTX 4080	CPU ~105W + GPU ~320W= ~425W	Liquid Cooling	High heat out -put, so liquid Cooling is better.
C. Intel core i3 + Integrated Graphics	CPU ~60-65 + Integrated graphics=low overall heat	Air-Cooling	Low heat output so air cooling is Enough.